Parametric vs. Non-Parametric Tests

1. Introduction

Statistical tests are broadly classified into **parametric** and **non-parametric** tests based on **assumptions** about the data.

Criteria	Parametric Test	Non-Parametric Test
Definition	Tests that assume specific **distributional properties** (usually normal distribution)	Tests that **do not require** assumptions about data distribution
Data Type	**Continuous (Numerical) Data** (e.g., height, weight, test scores)	**Ordinal or Categorical Data** (e.g., survey ranks, satisfaction levels)
Assumptions	- Data is **normally distributed** - **Homogeneity of variance** - **Equal sample sizes (sometimes)**	- No assumption about normality - Works with **skewed or ordinal** data
Examples	Z-test, t-test, ANOVA, Pearson correlation	Wilcoxon test, Mann-Whitney U test, Kruskal-Wallis test, Spearman correlation
When to Use?	If data follows a **normal distribution** and has **equal variances**	If data is **skewed, small sample size, ordinal, or non-normal**

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2. Parametric Tests

Parametric tests assume **specific conditions** about the population from which the sample is drawn.

Types of Parametric Tests

Test	Purpose	Example
Z-Test	Tests **mean difference** for **large samples** ($n \geq 30$) with known population standard deviation	Checking if the **average IQ = 100**
T-Test	Tests **mean difference** for **small samples** ($n < 30$) or unknown standard deviation	Comparing test scores of **two classes**
Paired T-Test	Compares **before and after** measurements for the same group	Checking if a **new drug reduces blood pressure**

Test	Purpose	Example
ANOVA (Analysis of Variance)	Compares **three or more group means**	Comparing **three teaching methods**
Pearson Correlation	Measures **linear relationship** between two continuous variables	Relationship between **study hours & exam scores**
Regression Analysis	Predicts the relationship between dependent and independent variables	**House price prediction based on size**

Example: T-Test (Parametric Test)

Problem: A company claims that its employees work an average of **40 hours per week**. A sample of **10 employees** shows an average of **38 hours** with a standard deviation of **4 hours**. At $\alpha = 0.05$, test if the claim is true.

Step 1: Define Hypotheses

- $H_0 : \mu = 40$ (no difference)
- $H_a : \mu \neq 40$ (working hours differ)

Step 2: Compute T-Statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{38 - 40}{4/\sqrt{10}}$$

$$t = \frac{-2}{1.26} = -1.58$$

Step 3: Compare with Critical Value

- For $\alpha = 0.05$, $t_{critical}$ (df=9) from the t-table = **±2.262**.
- Since $t = -1.58$ is within (-2.262, 2.262), we **fail to reject** H_0.

Conclusion

There is **no significant difference** in working hours.

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3. Non-Parametric Tests

Non-parametric tests **do not assume normality** and are useful for:

1. **Small sample sizes.**
2. **Ordinal or ranked data.**
3. **Skewed distributions.**
4. **Heterogeneous variances (unequal variances between groups).**

Types of Non-Parametric Tests

Test	Purpose	Parametric Equivalent	Example
Mann-Whitney U Test	Compares **two independent groups**	Independent **T-test**	Checking if **males and females have different test scores**
Wilcoxon Signed-Rank Test	Compares **paired data** (before/after)	Paired **T-test**	Checking if **a weight loss program is effective**
Kruskal-Wallis Test	Compares **three or more independent groups**	**ANOVA**	Comparing customer **satisfaction levels across 3 stores**
Spearman Correlation	Measures **rank-based relationship**	**Pearson Correlation**	Relationship between **customer rating & sales**
Chi-Square Test	Tests **associations between categorical variables**	None	Testing if **gender affects voting preference**

Example: Mann-Whitney U Test (Non-Parametric)

Problem: A company wants to check if **men and women have different salaries**. A sample of **10 men and 10 women** is collected. Since salary data is **not normally distributed**, we use a **Mann-Whitney U Test**.

Step 1: Define Hypotheses

- H_0: No difference in salary distribution between men and women.
- H_a: A difference exists.

Step 2: Rank Salaries

1. Rank all salaries (1 = lowest, 20 = highest).
2. Calculate the **U statistic**.

Step 3: Compare with Critical Value

- Using the **Mann-Whitney table**, if **U < critical value**, reject H_0.

Conclusion

If $p < 0.05$, we **reject** H_0 and conclude a **salary difference exists**.

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4. Comparison of Parametric vs. Non-Parametric Tests

Criteria	Parametric Tests	Non-Parametric Tests
Assumption	Requires normal distribution	No normality assumption
Data Type	Continuous (ratio, interval)	Ordinal, categorical, small samples
Efficiency	More powerful if assumptions are met	Less powerful, but robust
Examples	Z-test, t-test, ANOVA	Mann-Whitney U, Wilcoxon, Kruskal-Wallis
When to Use?	Large samples, normal distribution	Small, skewed, or categorical data

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5. When to Use Non-Parametric Tests?

1. **Data is not normally distributed** (skewed distribution).
2. **Sample size is small** ($n < 30$).
3. **Data is ordinal or ranked** (e.g., satisfaction levels).
4. **Comparing median instead of mean**.

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6. Summary Table

Scenario	Parametric Test	Non-Parametric Test
Comparing **two means**	T-Test	Mann-Whitney U Test
Comparing **three or more means**	ANOVA	Kruskal-Wallis Test

Scenario	Parametric Test	Non-Parametric Test
Comparing **paired data (before/after)**	Paired T-Test	Wilcoxon Signed-Rank Test
Checking **association between variables**	Pearson Correlation	Spearman Correlation
Checking **relationship between categorical variables**	None	Chi-Square Test